

```
import java.util.Scanner;

public class InsertionSort {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Get array size from user
        System.out.print("Enter number of elements: ");
        int n = scanner.nextInt();

        // Create and initialize an array
        int[] arr = new int[n];
        System.out.println("\nEnter element values:");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        // Sort the array using Insertion Sort
        insertionSort(arr);

        // Print the sorted array
        System.out.println("\nSorted array:");
        for (int i : arr) {
            System.out.print(i + " ");
        }
        System.out.println();
    }

    // Insertion Sort function
    private static void insertionSort(int[] arr) {
        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
```

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// Shift elements until key finds its correct position
while (j >= 0 && arr[j] > key) {
    arr[j + 1] = arr[j];
    j--;
}

// Place the key at its correct position
arr[j + 1] = key;
}
}
}
```